ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

MULTINOMIAL & DIRICHLET

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Inference with the Multinomial Distribution

Collection of proteins from the same family (similar sequence, structure, function). Each is a sequence of amino acids over a 20-letter alphabet $S = \{a_1, \dots, a_{20}\}$. You align them and you want to model the distribution over S at each position.

$$
\begin{array}{cccccc}
\cdots & a_1 & a_6 & a_7 & a_{20} & \cdots \\
\cdots & a_3 & a_8 & a_7 & a_8 & \cdots \\
\cdots & a_1 & a_9 & a_7 & a_{20} & \cdots
\end{array}
$$

Infer the distribution $\theta \in \Delta_{20}$ for the first position.

- Vector of counts at this position: $(x_1, \dots, x_{20}) = (2, 0, 1, 0, \dots, 0)$
- Here $(x_1, \dots, x_{20}) \sim$ multinomial$(n = 3, \theta)$

What is the maximum-likelihood estimate θ_{ML}?

Dirichlet Distribution

For $\alpha \in \mathbb{R}_+^k$ Dirichlet(α) is the distribution

$$\Pr(\theta_1, \dots, \theta_k) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \theta_1^{\alpha_1 - 1} \theta_2^{\alpha_2} \cdots \theta_k^{\alpha_k - 1}$$

over the probability simplex $\Delta_k = \{(\theta_1, \dots, \theta_k) : \theta_i \geq 0, \Sigma_i \theta_i = 1\}$.

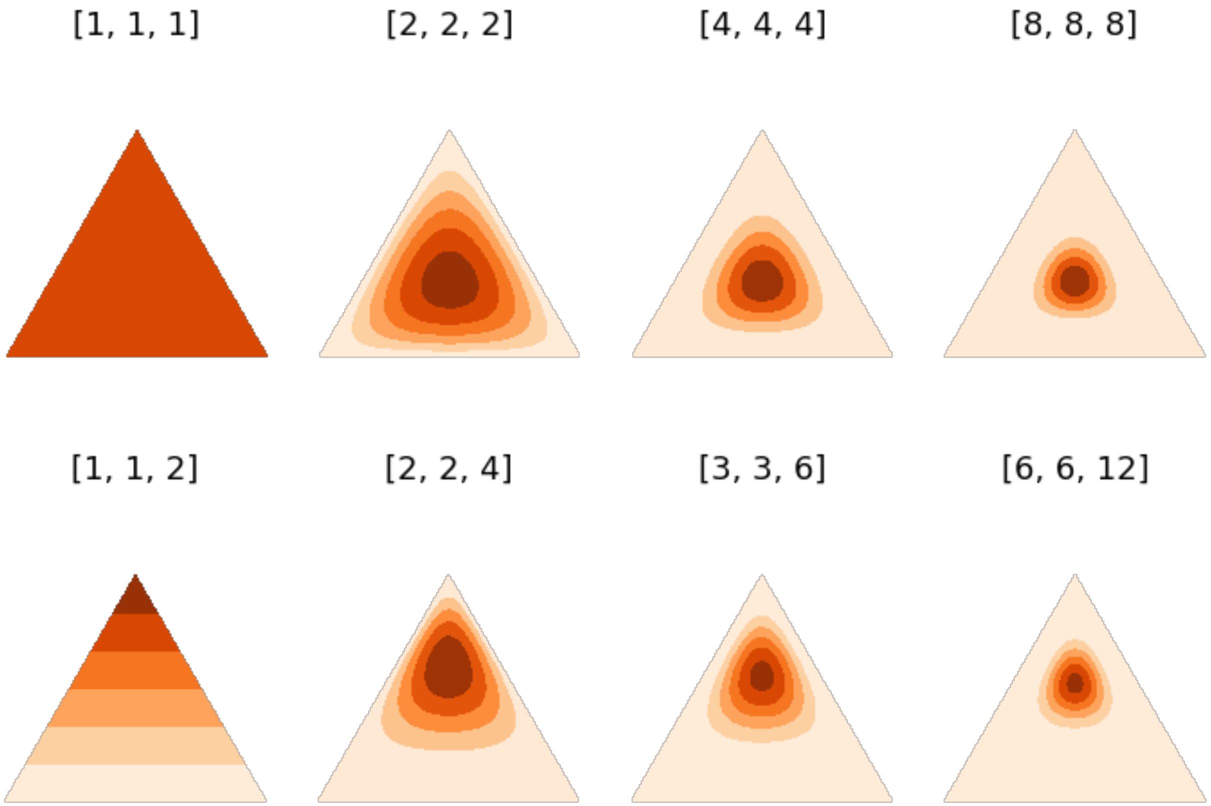
Properties:

$$\mathbb{E}\theta_i = \frac{\alpha_i}{\alpha_1 + \cdots + \alpha_k}$$

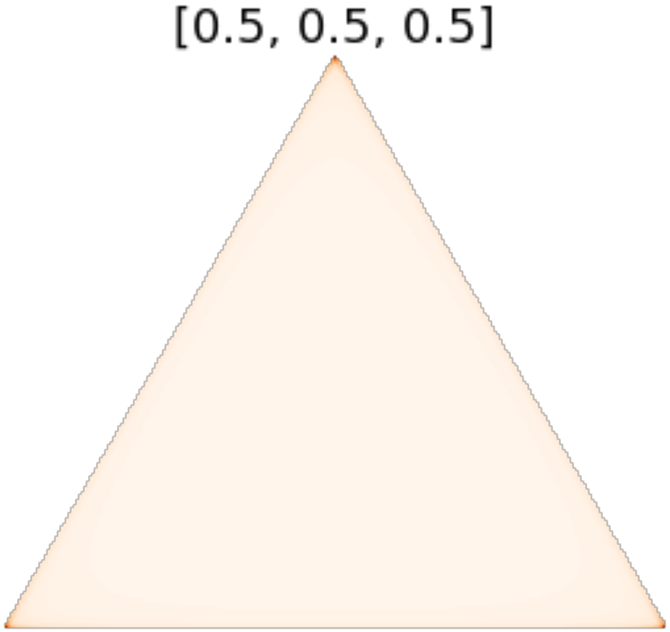
$$\text{Mode: } \theta_i = \frac{\alpha_i - 1}{\alpha_1 + \cdots + \alpha_k - k}$$

$$\mathrm{var}(\theta_i) = \frac{\alpha_i(\alpha_1 + \cdots + \alpha_k - \alpha_i)}{(\alpha_1 + \cdots + \alpha_k)^2(\alpha_1 + \cdots + \alpha_k + 1)}$$

Dirichlet Pictures



One More



Choosing the Prior Parameters

Bayesian inference for distribution $\theta \in \Delta_k$:

- Prior: Dirichlet$(\alpha_1, \dots, \alpha_k)$
- Draw n samples, get counts $(x_1, \dots, x_k)$

What is the posterior distribution?

Choosing the Prior Parameters

By analyzing databases of proteins, we find (hypothetically):

- 25% of positions are *highly conserved*: concentrated on a single amino acid.

$$a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}\ a_{11}\ a_{12}\ a_{13}\ a_{14}\ a_{15}\ a_{16}\ a_{17}\ a_{18}\ a_{19}\ a_{20}$$

- 12% of positions combine a particular set of amino acids with similar properties.

$$a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}\ a_{11}\ a_{12}\ a_{13}\ a_{14}\ a_{15}\ a_{16}\ a_{17}\ a_{18}\ a_{19}\ a_{20}$$

- 8% of positions combine a different set of amino acids.

$$a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}\ a_{11}\ a_{12}\ a_{13}\ a_{14}\ a_{15}\ a_{16}\ a_{17}\ a_{18}\ a_{19}\ a_{20}$$

But, how to combine these clusters?